

Computing in the Classroom: Assignment 01

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1 Introduction

In this paper, I will explore my portfolio as well as the pedagogical research that has been done and the impact it had on my design decisions. Inspired by Kate's fantasy world (where the VIKINGS create chaos across Scotland (Farrell 2024)), I created a game based off traditional algorithmic games with strong elements of audio dramas. The premise of the game is to retrieve the Honours of Scotland using a robot. The materials aims to address learning outcome 1-14a and 1-15a (EducationScotland 2024).

I developed the game using Twine (a tool for writing non-linear stories), on this platform learners will; listen and read the story, enter commands to the robot and solve simple puzzles. There is a copy of the map that is intended to be printed out, allowing learners to trace their commands to see any potential issues. Along side this, I created instructional documents enabling teachers to confidently use the materials independently.

2 Existing materials

Unplugged learning

We will now review existing materials that focuses on learning outcome 1-15a. The curriculum of excellence provides ideas about resources to use for each learning objective (EducationScotland 2024). For 1-15a, they suggest using physical robots such as BeeBots, Marty3 or Sphero. The main drawback of this is the associated cost. With prices ranging from £67.99 (tts 2024) to £350 (Robotical 2023), some schools are not able to feasibly buy physical robots for learners. This approach is a form of unplugged learning, which was found to have more significant positive effects in computational thinking than plugged-in learning (Sun, J. Liu, and Y. Liu 2024). In this study, Lihui found that both unplugged and plugged-in learning improves computational thinking over 10 weeks however, unplugged is more effective. The study focuses on Chinese learners Grades 1-2 (6-8 years old). Obviously, the Chinese educational system differs from the Scottish system. However, to test computational thinking, Lihui used Bebras problems, which are commonly found in the UK

school system. The study offers a comprehensive comparison of the two learning approaches. Given plugged-in learning can still improve computational thinking whilst being accessible to the majority of schools, I followed this approach to develop my materials. Another example of unplugged learning is the "Sending a rocket to Mars lesson"¹. This is a lesson plan created by CS Unplugged and is aimed to teach learners about the use of Algorithms. The lesson starts in the room and learners work in groups of three, where one learner creates the program, another reads out the program, and the last learner follows the commands. The aim is to guide "the bot" to a planet. This structure gives learners insights into all parts of the algorithmic process, and allows them to become a role they usually are not (program reader and program follower). The lesson then moves to a computer program where learners use a sequence of arrows to control a rocket. Later the learners are encouraged to evaluate and predict algorithms, as well as create algorithms for harder problems. These additional challenges are something I could consider implementing into my game as it allows all abilities to be challenged without seeming incomprehensible (I will touch on this in the research findings).

Plugged-in learning

Moving to plugged-in resources, they usually have a similar structure, Czakoova outlines some of the key elements of algorithmic games along with exploring games such as BeeBot emulator² (Czakoova 2020). Czakoova claims that educational games using basic control structures can allow learners to understand correctly how to get to the solution (Czakoova 2020). Games can achieve this by using; clearly defined steps with immediate feedback, a visualization tool for the sequence of steps, and opportunities for correction (Czakoova 2020). Unfortunately, there was no study conducted to support that these elements are important for improving algorithmic thinking. What we can take from this is that learners have a process of understanding algorithms as illustrated in figure 2, and implementing elements that help learners progress through this process will help understanding (Futschek and Moschitz 2010).

¹<https://www.csunplugged.org/en/topics/kidbots/sending-a-rocket-to-mars/>

²<https://beebot.terrapinlogo.com/>

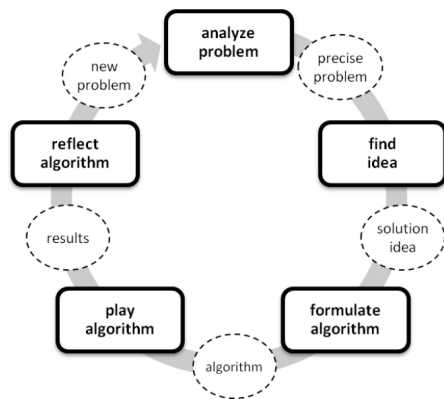


Figure 1: The process of learning by inventing algorithms (Futschek and Moschitz 2010)

The curriculum of excellence suggests using visual programming languages to meet 1-14a and 1-15a (EducationScotland 2024). There is an abundance of programs of this type but we will focus on Scratch³. One clear benefit of using Scratch is that learners get hands on experience. We will see later how this constructivist approach can positively impact learning. Most notably, Scratch has a low floor, high ceiling and wide walls (Maiorana 2019). This is ideal for learning environments, as learners at all stages can have room to grow. I wanted to provide learning materials that could be completed with no prior experience or the need for extensive set up for the teacher. There are examples of similar algorithmic games already available on Scratch, but they do not offer the immersive experience I aim to provide.

Notable resources

Finally, we will be discussing games where audio plays a key role. A resource that is already commonly used in UK schools is Kahoot!⁴. In Kahoot!, audio is played as the timer goes to zero as well as an indication if the player answered correctly or incorrectly. These sounds align with suggestions posed by Rovithis *et al.* (Rovithis et al. 2014). With ascending melodies for correct answers and descending melodies for incorrect answers. This might appear as a trivial design decision, however these background and feedback sounds have significant impact on learners (Wang and Lieberoth 2016). Wang conducted a study on the effect of points and audio on multiple factors⁵ when using Kahoot!. This detailed study with close to 600 students, found that there was a significant improvement to multiple factors when using audio. All the results apart from classroom dynamics were gathered through a questionnaire, it would have been more rigorous to use a formal testing system, although given the size of the group it would have been extremely difficult. The study was conducted on university students taking an IT introductory course, this is a very different demographic than my target age which might create some

inaccuracies when drawing conclusions for my target learner.

3 Research & Implementation

Constructivism

The game lends itself to a constructivist approach more specifically, Exogenous constructivism (Dalgarno 2001). This idea stems from the view that formal instruction, supplemented with exercises requiring learners to be cognitively active, can help learners form knowledge representations which they can later apply to other tasks (Dalgarno 2001). Dalgarno discusses the use of guided hypermedia; stating that these environments are designed specifically for learning discovery, encouraging the learner to follow certain instructional sequences but allowing them to choose alternative sequences (Dalgarno 2001). This is the inspiration behind the non-linear aspect of the game, as learners are given a problem (retrieving the Honours of Scotland), but are given the freedom to solve it how they see best. One criticism Dalgarno addresses of hypermedia is that learners tend to become “lost in hyperspace” (McKnight, Dillon and Richardson, 1991). One way to overcome this (as suggested by Dalgarno) is the use of an animated guide or intelligent agent (Dalgarno 2001). For this reason, when the learner has an opportunity to enter commands they are reminded where they are and which way they are facing (A appendix, figure 2). Additionally, if the learner goes over a trap/wall a personalised message will be displayed, informing them of what happened and where the robot will be returning to (A appendix, figure 3). This allows the learner to enter another command, whilst evaluating and improving on their previous attempt (A appendix, figure 2). This was seen as a key part of an effective algorithmic game (Czakoova 2020).

It was found that a constructivist approach has significant effect of the achievement of learners (Samaresh 2017). This study focused on Indian students of class 9th (around 14 years old), while studying physical science. The study was only conducted over two lessons which is not ideal for testing long term learning effects. However, with a sample size of 80 and all findings being a result of pre- and post-tests, the study has merit. They tested the 7E-model of constructivism, which aims to employ 7 E’s (Elicit, Engage, Explore, Explain, Elaborate, Evaluate and Expand) (Samaresh 2017). Most of these are naturally present in the game, and if learners are working in pairs then Explain and Evaluate will occur in conversation.

³<https://scratch.mit.edu/>

⁴<https://kahoot.com/>

⁵Concentration, Engagement, Enjoyment, Learning, Motivation, and Classroom Dynamics

Audial engagement

Another fundamental component of the game is the audio aspect. This is an area that has not been sufficiently researched, despite this there are insightful results that I have applied to my resources. The core reason for adding audio is to improve engagement. Following in alignment with Malone's theory of intrinsic motivation, an engaging activity should include; **challenge**, **fantasy**, and **curiosity** (Svendsen, Burner, and Røkenes 2020). The purpose of this is to create an activity that learners will engage with for its own sake and not an external reward (Svendsen, Burner, and Røkenes 2020). As Malone explains, external rewards have been observed to destroy intrinsic motivation particularly in children. To ensure the game has sufficient intrinsic motivation it is important to thoroughly consider how it brings; challenge, fantasy, and curiosity. The **challenge** is a direct result of the game premise, given learners limited experience controlling the robot will provide some challenge. Malone recommends adding in a variable difficulty level, this can be achieved through the puzzles. The learners will be able to select a difficulty that changes only the puzzles. This allows students to collaborate with moving through the lair together, while allowing them to solve problems that are catered more towards them. In the end I decided that it would be best if the puzzles were simple logic puzzles, as I wanted the game to last less than one hour. Malone mentions that feedback should be presented in a way that minimises the possibility of self-esteem damage (Svendsen, Burner, and Røkenes 2020). For this reason if the robot runs into a wall or trap, the I attempted to make the overall tone friendly (discussed more in Generative AI section). The **fantasy** is brought about from the audio descriptions throughout the story. These will emerge the learner, and allow for visualising the lair with details that are not included on the map. This is an example of intrinsic fantasy, although since learners are trying to retrieve the Honours of Scotland there is also extrinsic fantasies present (Svendsen, Burner, and Røkenes 2020). This goes hand-in-hand with the **curiosity** created in the game. Malone suggests the environment should be positioned between novel and surprising, without being incomprehensible. I have attempted to spark curiosity through audio as well as being set in twine which, almost magically, is in sync with the map.

Supporting this idea was a study conducted on the effects of game sound on learner engagement in digital game-based learning environments (Byun and Loh 2015). Byun *et al.* found that audial components of digital games are important factors affecting learners' engagement (Byun and Loh 2015). It is important to note that the mean age of the participants was 21 years with a sample size of 74 (52 males and 22 females). Differences in engagement tendencies between children and young adults will be vastly different however, this paper serves a specific purpose in demonstrating value

in voice-overs. Byun *et al.* carried out the test in a role playing game, where the player is given a set of task such as; visiting the market, talking to a non-player character (NPC), and getting a items from an NPC. The game I have developed is more puzzle focused, which could alter the effectiveness of voice-overs. Byun *et al.* make a case that engagement is on of the most important factors for successful learning. To collect results they used a modified Game Engagement Questionnaire (appendix A, figure 4). This addresses important aspects of a game which are highly relevant to my teaching materials. They found for 95% significance level on a forward stepwise regression model, the group an individual was in was the most significant factor (Byun and Loh 2015). Although for 90% significance level, prior RPG experience became a significant factor. This highlighted the importance of making the game accessible especially to those who have little experience in computer games.

Returning to the study on the effect of points and audio on Kahoot!, their results on engagement (A appendix, figure 5) describe the following. There was no statistically significant results from statement 3, but they found that those without audio and points were, to more of an extent, bored (16%) compared to the other groups (4-9%). The difference between audio and non-audio groups for statement 4 statistical significance, showing audio is a contributing factor to the feelings a learner experiences from a game. From this research I took that audio would be a key element of my game and implementing correctly would be vital. To do so, I added voice-overs for the introduction and the robot's narrative, as well as adding background and feedback sounds. This allows the learner to repeat the learning process by creating algorithms independently (figure 1).

4 Generative AI

I will now discuss the ways I used generative AI, and evaluate the responses generated. To begin with, I coded the game within Twine getting support from both online resources (documentation and videos) as well as Chat-GPT⁶. This was useful as I had never used Twine before and the available resources were quite limited. Chat-GPT handled simple tasks such as "I want the reader to only be able to make it to the next passage if they input the right value for a key". I soon reached a limit with what it was able to do correctly as tasks became more complex (creating a navigation system, adding audio, creating the typewriter effect). There exists two versions of Twine; Harlowe (the version I was using), and sugarcube. Chat-GPT would seem to prefer creating responses in sugarcube which are not compatible with Harlowe. These would often not be functional, but it was useful to see some potential macros that I could implement. This is how I created the navigation system, Chat-GPT gave me a

⁶<https://chatgpt.com/>

response that wouldn't function but made good use of the macros. I then adapted this to ensure it was running correctly. To create the token for SEARCH, I used Chat-GPT's image generation. It took a lot of tweaking, as it did not understand exactly what I wanted, but eventually I got something that I edited to get the final version. Being a language model, Chat-GPT is much better at writing than coding or image generation. For this reason I used it to generate the narrative of the robot. The responses would sometimes be convoluted and not fit the map that I created, so before adding them to Twine I edited them ensuring they were relevant. Finally, I used Consensus⁷ to aid in finding relevant papers. I found that Consensus would often exaggerate how impactful a study was, however it did provide useful examples that were related.

Conclusion

Hopefully I have illustrated how current materials surrounding algorithmic games and the research that has been done about the main aspects of my game, have influenced my design decisions. I recognise that it might have been more beneficial to have an updating map integrated into the game. Due to restrictions in Twine this is not feasible, but I believe the physical map has a lot to offer for learning experience. To improve on the existing game I could have included bonus missions such as what is the shortest path between two points, or complete part of the mission only using backwards moves and right turns. This could be independently added by the teacher, but it is something I wish I had time to properly implement.

Finally, I wanted to say thanks to Kate for helping me refine my ideas, as well as Jack McDermott who made a debut as the voice of SEARCH.

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⁷<https://consensus.app/search/>

Wang, Alf Inge and Andreas Lieberoth (2016). *The effect of points and audio on concentration, engagement, enjoyment, learning, motivation, and classroom dynamics using Kahoot!* URL: <https://www.semanticscholar.org/paper/The-effect-of-points-and-audio-on-concentration%2C-Wang-Lieberoth/1bcfb36bc724f12af7fc14824b2519adb819a27b>.

A Appendix

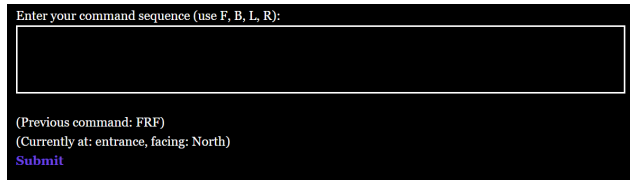


Figure 2: Sample of the game's interface showing; location, direction, and previous command

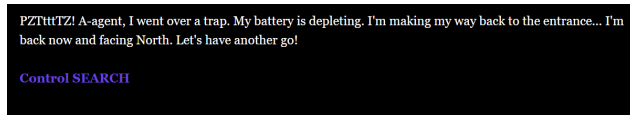


Figure 3: Sample of the game's interface, showing a message to the player

Items of the modified Game Engagement Questionnaire (mGEQ)

- | # | Item |
|-----|--|
| 1. | I lost track of time. |
| 2. | I was focused on the game rather than my surroundings. |
| 3. | I felt as if I was present at the High Harvest Fair. |
| 4. | I was disappointed when I had to stop playing the game. |
| 5* | I played without thinking about how to play. |
| 6. | Playing made me bored. |
| 7. | Time seemed to stand still or stop. |
| 8. | While playing, I thought about nothing but the game story. |
| 9. | I felt like I just couldn't stop playing. |
| 10. | I was curious to see how the game would end. |
| 11. | I really got into the game. |

* Removed due to a relatively low item-total correlation.

Figure 4: Modified Game Engagement Questionnaire (Byun and Loh 2015)

Statement	Group	Disagree	Agree	H	P
3. I thought playing the quiz was boring	Full Kahoot!	93%	7%	2.42	0.4899
	No audio	96%	4%		
	No points	91%	9%		
	No audio/points	84%	16%		
4. I felt increased pulse when answering questions	Full Kahoot!	31%	69%	50.56	<0.0001
	No audio	32%	68%		
	No points	61%	39%		
	No audio/points	68%	32%		

Figure 5: Results on engagement (Wang and Lieberoth 2016)